

Operator Discovery Report: candidate_pruner_scale_by_structure_escape_bias

- Operator type: candidate_pruner.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministic candidate-list size adaptation to improve transfer on clustered/grid/two-cluster-bottleneck cases; balances pruning aggressiveness via instance descriptors to reduce stagnation risk without exploding move e
- Novelty classification: descriptor_adaptive_pruner.
- Rediscovery signature: candidate_pruner:20:-2:-1:3.
- Surviving candidate: False.
- Problem class where it helps: none confirmed.

Pseudocode

- Start from a base candidate-list size.
- Expand or shrink the list using instance descriptors such as size, trap score, grid-likeness, clustering, and current stagnation.
- Run the downstream local search only on the resulting pruned neighborhood.

Complexity

- Non-empty lines: 12.
- Complexity score: 0.44.
- Fingerprint: 1c397e651ec235b4.

Why It Should Help

- The operator adapts neighborhood size to instance structure, which can trade off search breadth and runtime in a reusable way.

Failure Cases

- Underperforms on grid_like_tsp with gap delta 0.089548.
- Does not transplant cleanly into clustered_local_search.
- No validation family shows a clear improvement over the host baseline.
- Disabling the operator does not hurt, so the ablation does not support a real operator effect.

Transplant Results

- nearest_neighbor_2opt gap delta 0.000813, runtime delta 69.486486, distance-eval delta -374.857143.
- cheapest_insertion_2opt gap delta 0.0, runtime delta 740.939957, distance-eval delta 62.285714.
- random_restart_2opt gap delta -0.000421, runtime delta -48.754557, distance-eval delta -493.285714.
- sparse_three_opt gap delta 0.000379, runtime delta 20.086271, distance-eval delta 1797.857143.
- clustered_local_search gap delta 0.025002, runtime delta -118.296786, distance-eval delta -3584.285714.

Instance-Family Results

- heldout_tsplib operator gap 0.235871 vs baseline 0.235058.
- clustered_tsp operator gap 0.110853 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.196244 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.109057 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: 0.009726.
- Runtime inflation: -0.126442.
- Distance-evaluation inflation: -0.01997.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator -0.009726.
- constant_pruner mean gap 0.121317 and delta vs full operator -0.009726.
- wide_pruner mean gap 0.121542 and delta vs full operator -0.009501.